

Fine-tuning a Tsuboyama-pretrained $\Delta\Delta G$ model on FireProt

Experiment 08 · ddG_with_Boltz · concat features + antisymmetry · MLP · cross-dataset homology holdout.

A $\Delta\Delta G$ regressor pretrained on Tsuboyama and then **sequentially fine-tuned on FireProt** is the **only** configuration that is good on both datasets: it beats both a Tsuboyama-only and a FireProt-only baseline on FireProt-test (at 30/50 % identity, and on Spearman throughout) while losing essentially nothing on Tsuboyama (≤ 0.012 Pearson). Training on FireProt alone matches the transfer on FireProt but collapses on Tsuboyama — so fine-tuning genuinely combines the two datasets rather than trading one for the other. The gain is modest, bounded by the feature ceiling.

1. Motivation

A Tsuboyama-trained model transfers to the independently curated FireProt dataset but is bounded by how densely the training data covers each $\Delta\Delta G$ value. A natural way to add coverage is to fine-tune on FireProt itself. The question is whether this helps FireProt, and whether it degrades the original Tsuboyama performance (catastrophic forgetting) — measured on both datasets under a split that controls for homology across them.

2. Methods

Splits. All wild-type sequences from both datasets are pooled and clustered by sequence identity (MMseqs2, 80% coverage) at 30/50/90%. Whole clusters are assigned to train or test, so no train/test pair — within or across datasets — exceeds the threshold; clusters spanning both datasets go to train. This yields disjoint *Tsuboyama-train* / *Tsuboyama-test* / *FireProt-finetune* / *FireProt-test* sets.

Model. 5-seed MLP ensemble on concat pair-track features with antisymmetry augmentation (the defaults established in experiment 07), evaluated in three conditions. **A (Tsuboyama-only):** trained on Tsuboyama-train. **B (FireProt-only):** a fresh model trained on FireProt-finetune alone, with its own input normalisation — the baseline for "how far does FireProt get on its own." **D (fine-tuned):** the pretrained model, warm-start continued on FireProt-finetune at a lower learning rate, reusing the Tsuboyama normalisation. All three are evaluated on Tsuboyama-test and FireProt-test.

3. Results

Identity	Pearson A	Pearson B	Pearson D	Spearman A / B / D
30%	0.466	0.477	0.522	0.693 / 0.661 / 0.739
50%	0.507	0.505	0.528	0.672 / 0.703 / 0.716
90%	0.355	0.291	0.343	0.658 / 0.634 / 0.686

Table 1. FireProt-test pooled correlation, A / B / D. Bold = best (D).

Identity	A (Tsu-only)	B (FP-only)	D (fine-tuned)
30%	0.794	0.680	0.790
50%	0.787	0.692	0.784
90%	0.790	0.678	0.778

Table 2. Tsuboyama-test pooled Pearson r, A / B / D (forgetting check).

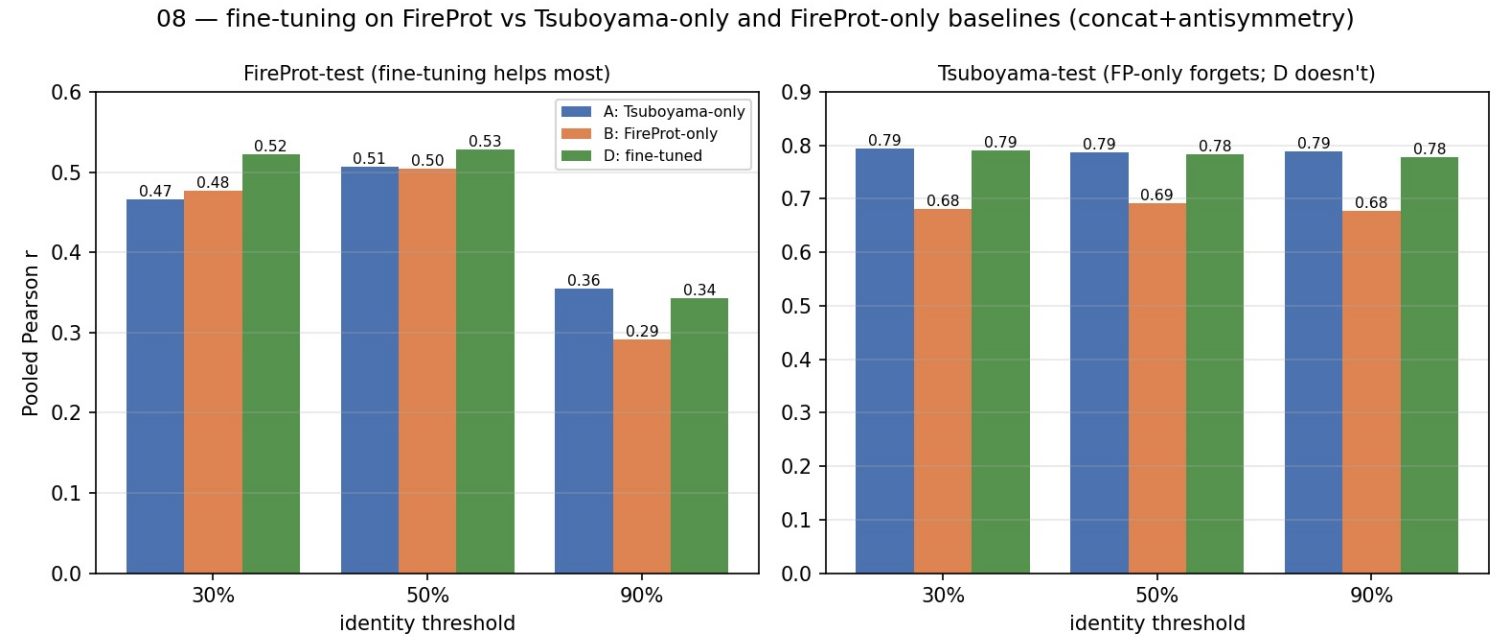


Figure 1. Pooled Pearson r for A / B / D. Left: FireProt-test (D best at 30/50%). Right: Tsuboyama-test — FireProt-only (B) drops to ~0.68 while A and D stay ~0.79 (D does not forget).

4. Discussion

The fine-tuned model (D) is the **only condition that is strong on both datasets**. On FireProt-test it beats both baselines at 30% and 50% identity and improves Spearman at all three thresholds; on Tsuboyama-test it is within ≤ 0.012 of the Tsuboyama-only model — no meaningful forgetting. The FireProt-only baseline (B) is instructive: it matches the transfer *on FireProt*, but its Tsuboyama-test correlation collapses from ~ 0.79 to ~ 0.68 — training on the small FireProt set alone discards the broad Tsuboyama signal. Fine-tuning therefore *combines* the two datasets rather than trading one for the other. The overall size of the gain is modest — the predictor's accuracy is set largely by the pair-track features, so adding FireProt labels refines rather than transforms it. At the strictest 90% split the FireProt-test set is smallest and noisiest (231 mutations), where Pearson dips while Spearman still rises — a calibration wobble, not a loss of ranking. Sequential fine-tuning is a safe, mild improvement when a second labelled dataset is available.